THE DISTINGUISHED TRIANGLE FOR A COMPOSITION OF MORPHISMS OF RINGED TOPOI

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement displays the triangle; the proof is short and consists of the identification of the terms and the appeal to the sheaf-theoretic triangle lemma.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The pullback description of the cotangent complex along a morphism of ringed topoi. Lemma lemma-pullback-cotangent-morphism-topoi, used at proof lines 3958-3961. Identifies f^{-1}L_g with a cotangent complex of sheaves of rings.
- The distinguished triangle for a composition of sheaves of rings. Lemma lemma-triangle-sheaves-rings (3494-3505), used at proof lines 3962-3966. Supplies the triangle with A = h^{-1}O_3, B = f^{-1}O_2, C = O_1.
- Flatness of the complex identified with f^{-1}L_g. used at proof lines 3960-3961. Makes the triangle lemma applicable in the form stated.

Source reference. Original lemma 08SV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11143–11143. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08SZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11147–11147. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $f : (Sh(\mathcal{C}_1), \mathcal{O}_1) \to (Sh(\mathcal{C}_2), \mathcal{O}_2)$ and $g : (Sh(\mathcal{C}_2), \mathcal{O}_2) \to (Sh(\mathcal{C}_3), \mathcal{O}_3)$ be morphisms of ringed topoi. Then there is a canonical distinguished triangle*

$$Lf^*L_g \to L_{g\circ f} \to L_f \to Lf^*L_g[1]$$

in $D(\mathcal{O}_1)$.

Proof. Set $h = g \circ f$ so that $h^{-1}\mathcal{O}_3 = f^{-1}g^{-1}\mathcal{O}_3$. By Lemma 08SV we have $f^{-1}L_g = L_{f^{-1}\mathcal{O}_2/h^{-1}\mathcal{O}_3}$ and this is a complex of flat $f^{-1}\mathcal{O}_2$-modules. Hence the

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distinguished triangle above is an example of the distinguished triangle of Lemma 08SZ with $\mathcal{A} = h^{-1}\mathcal{O}_3$, $\mathcal{B} = f^{-1}\mathcal{O}_2$, and $\mathcal{C} = \mathcal{O}_1$. $\square$